EE/CprE/SE 491 WEEKLY REPORT 1

Feb 17 20205 – Feb 25 20205

Group number: 16

Project title: Outdoor Roomba

Client &/Advisor: Prof. Diane Rover

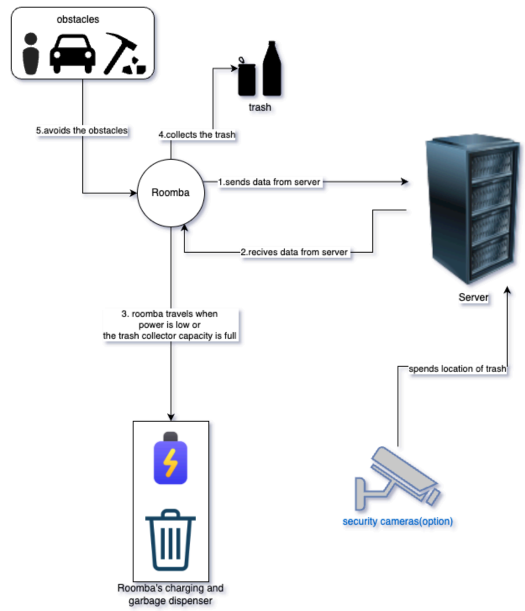
Team Members/Role: Brodie M Bates, Daniel Vergara Pinilla ,Mitchell D Owen,Om Shukla, Sudipta Halder

- **Weekly Summary**

Our goal for this week was to start planning on the Roomba and scope down to the requirements so we can finish this project by December 2025.First we determined a time to meet our advisors and a meeting time for the team itself. Then we had our first meeting with our advisors. We researched and brainstormed ideas for the garage-picking mechanism. We picked a few design ideas for prototyping. Then, we added some requirements that benefit the users. For Instance, insurance doesn't cover damage created by a robot, so we need the Roomba to be smart enough not to cause damage and be physically harmless so it doesn't run over people or pets. Lastly, we ordered a VM from ETG, which we will use as a server.

- ## **Past week accomplishments**

Top Level Diagram of the Project



- Team Member 1 (Sudipta Halder): This week, we worked on setting up a Discord Server so our team could brainstorm and research different channels. Started brainstorming with the team. Decided on a testing ground for the Roomba to be inside the courtyard of Coover. Then worked on creating a top-level diagram of our project. Ordered an Ubuntu Virtual machine with 8 GB of RAM and 50 GB of storage from ETG with the hostname of SDDEC25-16.

- Team Member 2 (Brodie Bates): This week, researched other robots and how they would pick up items. Alongside, drew two designs of our lifting mechanism with the robot base. Also took notes during our advisor meetings.

- Team Member 3 (Daniel Pinilla): This week, worked on making a 3D design of a possible prototype option. As well as working on testing the functionality of the Coral AI chip

- Team Member 4 (Om Shukla): This week, worked on looking at how other robots of a similar type are programmed and what they have in functionality. Why those products succeeded and what sets them apart from others.

- Team Member 5 (Mitchell Owen): This week, worked on doing research on their products out there. My two main autonomous robots, focused on were other garbage picking up robots and lawnmower robots. Wanted to look at lawnmower robots because they deal with some harsher conditions and we need our robot to be able to handle the same if not worse conditions. For trash picking up robots that we can compare to; found the Trombia which is close to our vision for our robot.

- **<u>Pending issues</u>** *(If applicable: Were there any unexpected complications? Please elaborate.)*
  - Team Member 1 (Sudipta Halder): One of the biggest problems is that we are not going to be in the same room at the same time since our schedules are very different from each other. So we need to work asynchronously on a Hardware Problem which usually requires a lot of in-person working hours.Little worried about getting the team together to work on things.

  - Team Member 2 (Brodie Bates): No pending issues

  - Team Member 3 (Daniel Pinilla): My biggest fear is that everyone agrees on all the ideas, which make it hard sometimes to make final decisions because we still have many options agreed

  - Team Member 4 (Om Shukla): No pending issues

  - Team Member 5 (Mitchell Owen): The biggest issue, have is that we don't do a lot of in-person meetings.Hopefully that this gets better with the weather getting nicer, but currently, the motivation is lacking more when we are online having the meetings. Another issue with this is that we don't believe we are always on the same page. Keep asking questions about what our plans are because we don't understand the idea, but luckily we are really good at asking questions and we are just as good at clarifying.

- **Individual contributions** *(Creating this section is optional, but it is **Required to include the "Hours Worked for the Week" and their "Total Cumulative Hours" for the project for each member somewhere relevant in your report. Your individual weekly hours should be at a minimum of 6-8 hours for this course. So please manage your time well. Also, ensure that individual contributions support your claim to the weekly hours. Be honest with the reports.**)*

NAME	**Individual Contributions** *(Quick list of contributions. This should be short.)*	**Hours this week**	**HOURS cumulative**
Brodie Bates	Designs, Research, Notes	6	6
Om Shukla	Research	6	6
Sudipta Halder	Setup meeting and Discord, Research, and training	5	5
Daniel Mauricio Vergara	3D Design of possible prototype and ML initial testing	10	10
Mitchell Owen	Research	6	6

- **Plans for the upcoming week** *(Please describe duties for the upcoming week for each member. What is(are) the task(s)?, Who will contribute to it? Be as concise as possible.)*

  - Team Member 1 (Sudipta Halder): Will work on prototyping the roomba by designing it in fusion 360 and if we are done making an early build of it we will go on to build it. Also Try to create a clear sprint diagram so our team knows what to do next.
  - Team Member 2 (Brodie Bates): Next week will be going to work on prototyping some of the mechanisms for our robot. As well as researching some of our other requirements such as weatherproofing.
  - Team Member 3 (Daniel Pinilla):Next week I'll work on implementing a wheel design for the prototype as well as starting testing on the raspberry pi
  - Team Member 4 (Om Shukla): Next week, planning to get started on researching how the Roomba needs to communicate with the server for information and when the communications will happen. Also want to research how other companies have created their products and what kind of server communication they require, if any.
  - Team Member 5 (Mitchell Owen): My plan is to find ways to integrate current products that are out there with ours so we can find a way to make it better while trying to stay within budget.

- **<u>Summary of weekly advisor meeting</u>**
  During our advisor meeting we discussed specific users and use cases for our outdoor roomba as well as what type of environment we intend for the roomba to be used in. We were told to check out some of the products on the market that are similar to our project and take inspiration from them. We decided to have a test area in a controlled environment before we move to testing outside. We started making a list of requirements that we can realistically achieve.